Robust Impact Mitigation with Heterogeneous Nonlinear Metamaterials: Replicating and Extending a Discrete-Element Study

Checkpoint Report

[**TODO:** author names]

November 7, 2026

Abstract

Nonlinear elastic constitutive laws, in particular bistable and snap-through springs, can strongly attenuate the kinetic energy that an impact transmits through an architected material, but optimized nonlinear designs are known to be highly sensitive to small changes in material parameters and impact conditions. Tamur *et al.* [1] showed computationally that making the *nonlinear* part of the spring laws spatially heterogeneous improves this robustness. In this project we replicate their study and use it as a testbed for data-driven design. We re-implemented their one-dimensional discrete element model (DEM): a 20-mass chain with four families of cubic-polynomial springs, struck by a light impactor. The re-implementation is a compiled, batched Python solver that matches the authors' MATLAB code to a median relative difference of 6×10^{-7} in the performance metric and runs about 500 simulations per second on four CPU cores. With it we reproduced the paper's impactor-mass perturbation results (to two significant figures) and its coefficient-noise robustness trend. We also generated a dataset of 360,000 simulated designs that spans the paper's admissible design space. Preliminary sensitivity analyses confirm the paper's central qualitative finding: homogeneous bistable chains are the most sensitive to perturbations and heterogeneous chains without bistable springs are the least. The next stage replaces the gradient-based inverse design of the original work with Gaussian-process (GP) surrogate optimization.

1 Introduction

1.1 Background and problem statement

Protecting a structure or a person from impact means limiting how much of the incoming energy reaches the protected side. Material heterogeneity is known to raise damage tolerance in quasi-static settings, as in bone and other biological materials [2, 3]. Separately, in dynamic settings, nonlinear elastic responses have emerged as a passive way to reshape stress waves [4, 5]. Strong nonlinearity combined with dispersion can break an impulse into a train of smaller solitary waves [5], and bistable elements can trap strain energy or dissipate it [6, 7]. The drawback is sensitivity. Nonlinear systems can respond very differently to small changes in impactor condition or material properties [8], and bistable chains can even localize and transmit *more* kinetic energy than their linear counterparts [9]. A design that is optimal on paper may therefore perform poorly once manufactured.

We study the model of Tamur *et al.* [1]. A chain of $N = 20$ unit masses is joined by springs that are linear in tension and cubic in compression, each with a viscous damper in parallel ($\zeta = 0.02$),

and fixed to a wall at the far end. In compression the spring law of family I is

$$f_I(\Delta x) = \Delta x + c_{2I}\,\Delta x^2 + c_{3I}\,\Delta x^3, \qquad I \in \{A, B, C, D\}, \tag{1}$$

with five consecutive springs per family, so that the linear stiffness is uniform and only the nonlinear terms vary along the chain. A point-mass impactor ($M/M_0 = 0.01$, $V/V_0 = 0.25$, with M_0 half the chain mass and V_0 the linear sound speed) hits the first mass through a tension-free Hertzian contact $f_H = C_{\text{imp}}[x_0 - x_1]_+^{3/2}$ with $C_{\text{imp}} = 1.2 \times 10^4$. The performance metric, which is also the design objective, is the *KE ratio*

$$\text{KE} = \frac{\max_t \text{KE}_N^{\text{non}}(t)}{\max_t \text{KE}_N^{\text{lin}}(t)}, \tag{2}$$

the peak kinetic energy of the last mass, divided by the same quantity for the purely linear chain. Lower is better. Admissible designs satisfy $c_2 \in [-120, 0]$, $c_3 \in [0, 4000]$ and positive strain energy, $c_3 > \frac{2}{9}c_2^2$, for every family. The roots of f and f' split this space into monotonic ($c_2^2 \le 3c_3$), snap-through ($3c_3 < c_2^2 \le 4c_3$) and bistable ($c_2^2 > 4c_3$) springs [1].

Three design categories are compared: *homogeneous* (one spring law, 2 variables), *heterogeneous* (four laws, 8 variables) and *heterogeneous excluding bistability* (four laws, with $c_2^2 \le 4c_3$ in every family). **Our problem** is twofold: (i) to reproduce, independently, the paper's evidence that heterogeneity in the nonlinear constitutive response improves robustness to uncertainty in the spring coefficients and impact conditions; and (ii) to evaluate a sample-efficient, data-driven alternative to the paper's finite-difference gradient optimizer, namely GP surrogate optimization, on this expensive and rugged objective.

1.2 Literature review and previous work

Disorder and wave transport. Anderson showed that sufficient disorder can stop energy from diffusing in linear lattices [10]. Related ideas have been used to widen band gaps in phononic crystals and acoustic metamaterials [11]. In nonlinear lattices the effect of disorder is less clear: depending on the case, nonlinearity has been found to preserve, destroy or modify localization [12].

Nonlinear and bistable metamaterials for impact. Granular and other strongly nonlinear chains convert impulses into solitary waves [5, 4]. Multistable architected materials can trap elastic strain energy [6]. In bistable chains, the absorbed kinetic energy depends non-monotonically on impactor mass and velocity [9]. Reducing intersite viscosity can increase relative energy absorption, but only when the chain is finely enough discretized [7]. MacNider *et al.* used bilevel inverse design to tailor nonlinear constitutive laws for target wave responses, and found that optimal homogeneous designs were bistable [13].

The replicated study. Tamur *et al.* [1] optimized the coefficients (c_2, c_3) of Eq. (1) with MATLAB's `fmincon` (finite-difference gradients, random initial designs) and integrated the DEM with `ode45` [14, 15]. The best KE ratios they report are 0.56% (heterogeneous), 1.17% (homogeneous) and 0.86% (heterogeneous excluding bistability). They then perturbed the coefficients by Gaussian noise of level $\epsilon \le 5\%$. At $\epsilon = 5\%$, the homogeneous design's KE ratio worsened 8×, the heterogeneous design's 12× (from a lower baseline), and the bistability-free design's only 3×. Over ensembles of 200 optimized designs per category, the median perturbed KE ratio was 0.105, 0.067 and 0.050 respectively.

Surrogate-based optimization. Every KE evaluation requires two stiff, long-time ODE integrations, and the objective landscape is rugged (Fig. 2). Gaussian-process regression [16] with expected-improvement acquisition [17] is a standard way to optimize such expensive black-box functions with few evaluations. GP models also provide predictive uncertainty, which is directly relevant to a robustness study.

1.3 Description of the data

All data are generated by simulation; no experimental measurements are involved.

Simulator. We ported the authors' MATLAB DEM to Python twice. The first port is a line-by-line `scipy` version [18] that serves as a reference. The second is a Numba-compiled [19] adaptive Dormand–Prince 5(4) integrator (the same method as `ode45` and `scipy`'s RK45) that runs one simulation per CPU thread, stops at $T = 25/f_0 \approx 497$, and takes the peak of KE_N continuously from the solver's dense output. Tolerances are rtol = atol = 10^{-12}. Table 1 summarizes the verification.

Dataset. The dataset contains 360,000 nominal designs, 120 000 per category, drawn uniformly from the admissible region by rejection sampling with seed 0. Each row holds:

- **inputs**: $c_2, c_3 \in \mathbb{R}^4$ for families A–D (8 features; for homogeneous rows all four families share one pair);
- **target**: the KE ratio (Eq. (2)); it spans more than three decades (Fig. 4), so we model $\log \mathrm{KE}$;
- **metadata**: category, the SI-3 spring type of each family, solver step count and status. All 360,000 integrations succeeded.

Across all families, 84.3% of the sampled springs are monotonic, 12.6% snap-through and 3.1% bistable. Good designs are rare: only 92, 405 and 317 designs (homogeneous, heterogeneous, excl. bistability) reach $\mathrm{KE} \leq 0.03$. The best random designs reach 0.0114, 0.0089 and 0.0112, versus the paper's optimized 0.0117, 0.0056 and 0.0086. The generator can also attach Monte Carlo perturbed copies of each design (paper Eq. 4), which provides robustness labels (mean and standard deviation of KE under noise).

2 Preliminary analysis

Simulator verification. Table 1 lists the checks that pass. All of them are reproducible with the script `check_simulation.py` in the `fast` folder. The equations of motion agree with MATLAB to machine precision. The KE ratio agrees with MATLAB's `ode45` to a median relative difference of 6×10^{-7}. For the paper's Fig. 4 case (impactor mass increased by 1%), we reproduce every reported number. Four of the six designs tabulated in the paper's SI agree to within 3%. The remaining two (homogeneous optimum and heterogeneous perturbed) contain bistable springs. Perturbing their published coefficients by less than the ± 0.005 rounding of the table moves KE over 0.0117–0.062 and 0.0117–0.038, and both ranges contain the paper's values. We therefore attribute the mismatch to the printed precision, not to the model. Measuring the peak on the paper's coarser output grid instead changes KE by less than 0.3%.

Table 1: Verification of the Python DEM. Paper values are from Tamur *et al.* [1] (SI Tables S1–S2 and Sec. 3).

Check	Ours	Reference
Equations of motion, 5000 random states	max diff. 6×10^{-16}	MATLAB
KE ratio, 16 designs	median rel. diff. 6×10^{-7}	MATLAB `ode45`
Heterogeneous optimum	0.0057	0.0056
Het. excl. bistability optimum	0.0086	0.0086
Homogeneous, perturbed	0.0627	0.062
Het. excl. bistability, perturbed	0.0145	0.015
Linear chain, $1.01M$: peak-KE increase	$\times 1.018$	$\times 1.02$
$1.01M$: KE homo / hetero / excl.	0.109 / 0.026 / 0.0077	0.11 / 0.026 / 0.008

One-at-a-time coefficient sensitivity. Figure 1 varies each coefficient of the example design from the authors' code ($c_2 = [-70, -100, -5, -35]$, $c_3 = [1200, 2000, 1500, 3000]$, KE = 0.148) by $\pm 50\%$, one at a time. The two families nearest the impactor (A, B) dominate: changing their c_2 alone moves KE between 0.047 and 2.7, while families C and D barely matter. Lowering c_3 in A or B quickly drives the spring out of the positive-energy region (hollow markers), where KE exceeds the linear chain by up to $6.4\times$. The response is also strongly non-monotonic, which already signals a rugged objective.

Sensitivity to impact conditions. Figure 2 slices KE around the design point for the three optimized designs of the paper (paper Fig. 2b). The homogeneous bistable chain is extremely jagged. Its mean $|\partial \mathrm{KE}/\partial(M/M_0)|$ is 1405, against 221 (heterogeneous) and 42 (excl. bistability). The corresponding values for $|\partial \mathrm{KE}/\partial(V/V_0)|$ are 56, 9.3 and 3.1. This reproduces the paper's ordering and magnitudes (its Fig. 2c).

Sensitivity to coefficient noise. Figure 3a repeats the paper's Fig. 1c with 500 perturbed samples per noise level, $\epsilon \in [0, 5\%]$. At $\epsilon = 0.25\%$ the mean KE is already 0.071 (homogeneous), 0.015 (heterogeneous) and 0.0091 (excl. bistability). At $\epsilon = 5\%$ it is 0.087, 0.074 and 0.025, close to the paper's curves. With the paper's unrounded homogeneous optimum (0.0117) these are degradations of roughly $6\times$, $3\times$, $1\times$ and $7.5\times$, $13\times$, $3\times$, compared with the reported $6\times, 3\times, 1\times$ and $8\times, 12\times, 3\times$. The ordering at 5% noise is the paper's: excluding bistability is clearly the most robust.

Category-level robustness (preliminary). Following the paper's Fig. 1d protocol, we took designs from our random dataset with nominal KE $\in [0.01, 0.03]$, balanced over four bins (20, 84 and 60 designs; too few random designs reach the window to obtain the paper's 200), and ran 100 Monte Carlo trials at $\epsilon = 5\%$ (Fig. 3b). The median of the per-design mean μ_{KE} is 0.107 (homogeneous), 0.046 (heterogeneous) and 0.031 (excl. bistability), against the paper's 0.105, 0.067 and 0.050. The fraction with $\mu_{\mathrm{KE}} < 0.03$ is 0%, 23% and 47% (paper: 0%, 0%, 9%). The category ordering and the homogeneous value match. The heterogeneous sets are more robust than in the paper, which we expect, because random designs that land in the window are not the sharp optima produced by an optimizer. Building optimized baseline sets is the motivation for the GP stage.

Data characteristics relevant to learning. The target is heavy-tailed and multimodal (Fig. 4). The best designs lie in a thin tail ($< 0.3\%$ of samples below 0.03), and the response is jagged at small scales (Figs. 1 and 2). For a GP surrogate this suggests modeling $\log \mathrm{KE}$, using a stationary kernel with short length scales or a noise term to absorb sub-resolution roughness, and treating the snap-through/bistable boundaries as potential discontinuities.

3 Figures

3.1 Sensitivity analysis

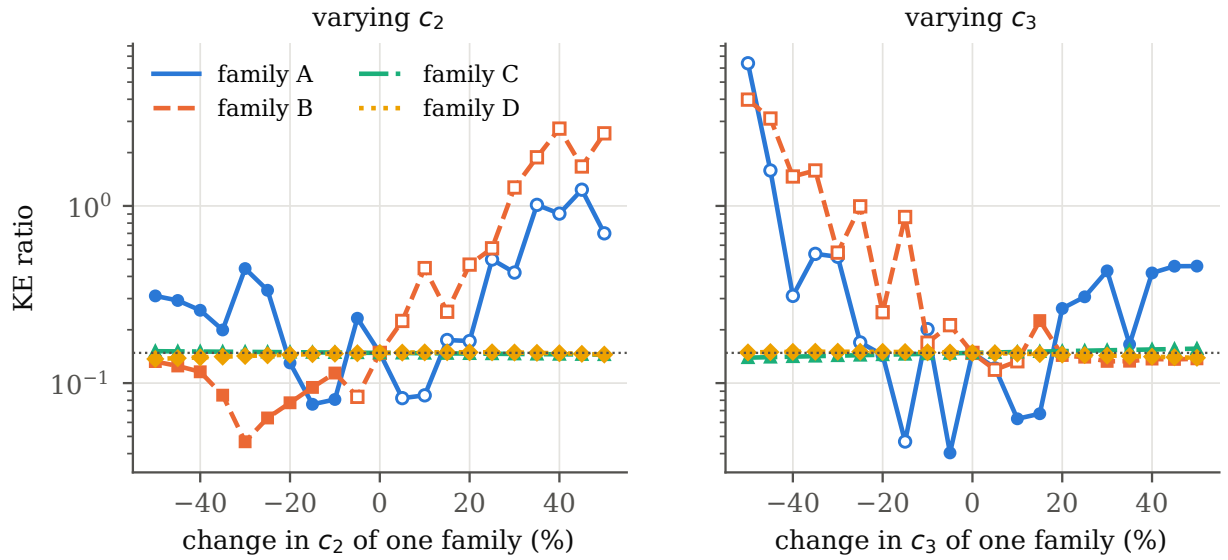


Figure 1: One-at-a-time sensitivity of the KE ratio to c_2 (left) and c_3 (right) of each spring family, $\pm 50\%$ around the example design (dotted line, $\mathrm{KE} = 0.148$). Hollow markers violate the positive-energy constraint $c_3 > \frac{2}{9}c_2^2$. Log scale.

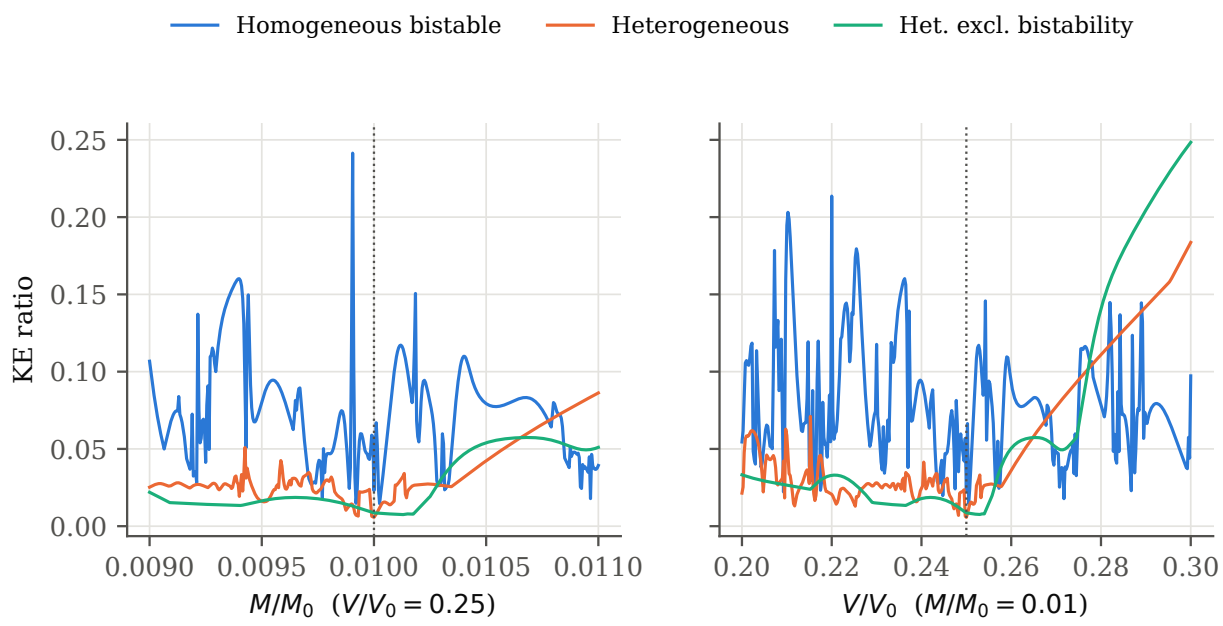
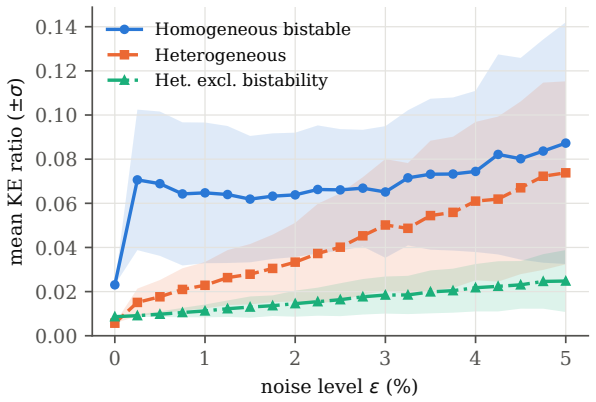
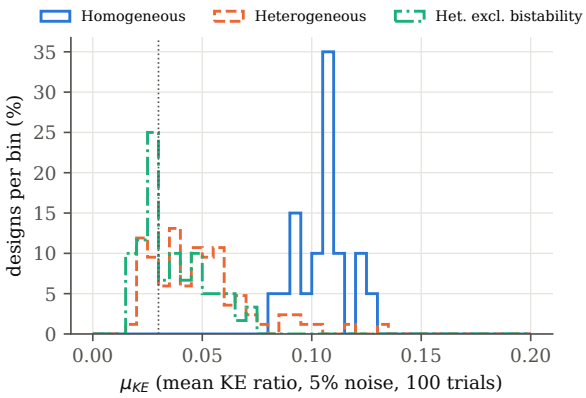


Figure 2: KE ratio of the paper's three optimized designs along slices in impactor mass (left) and velocity (right) through the design point (dotted). Compare with Tamur *et al.* Fig. 2b.



(a) Mean ± one standard deviation of the KE ratio under Gaussian coefficient noise of level ϵ (500 samples per level). Compare with paper Fig. 1c.



(b) Distribution of μ_{KE} for random designs with nominal KE $\in [0.01, 0.03]$ after 5% noise (100 trials each); dotted line at 0.03. Compare with paper Fig. 1d.

Figure 3: Robustness to uncertainty in the nonlinear spring coefficients.

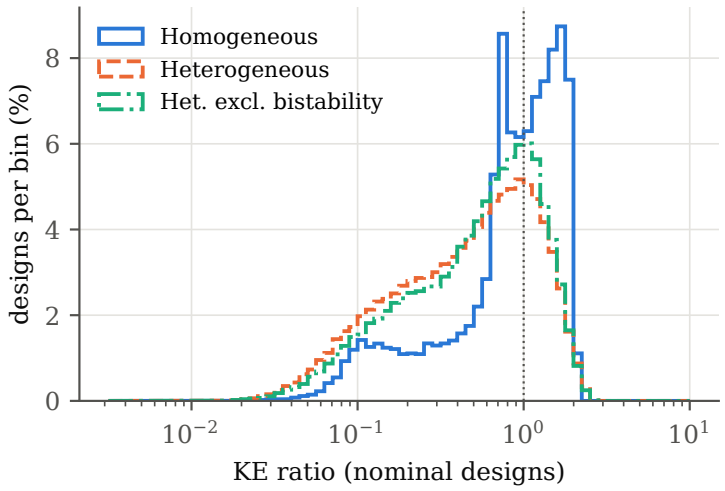


Figure 4: Distribution of the KE ratio over the 360 000-design dataset, by category. Designs left of the dotted line outperform the linear chain.

3.2 GP optimization results

Placeholder: GP optimization not yet run.

Planned panels: (a) best KE found vs. number of simulations, GP-EI vs. random search vs. finite-difference gradient optimization, per category;
(b) GP predicted vs. simulated log KE on held-out designs;
(c) optimized (c_2, c_3) over the SI-3 category boundaries (paper Fig. 1e).

Figure 5: [**TODO:** GP surrogate optimization results (next milestone).]

References

[1] C. Tamur, M. Brandy, H. A. Kim, and N. Boechler. Heterogeneity in constitutive nonlinearity enhances impact mitigation robustness. Manuscript and Supplementary Information.

[2] T. M. Keaveny and W. C. Hayes. A 20-year perspective on the mechanical properties of trabecular bone, 1993.

[3] U. G. K. Wegst, H. Bai, E. Saiz, A. P. Tomsia, and R. O. Ritchie. Bioinspired structural materials. *Nature Materials*, 14(1):23–36, 2015.

[4] G. U. Patil and K. H. Matlack. Review of exploiting nonlinearity in phononic materials to enable nonlinear wave responses. *Acta Mechanica*, 233(1):1–46, 2022.

[5] V. F. Nesterenko. *Dynamics of Heterogeneous Materials*. Springer, New York, 2001.

[6] S. Shan, S. H. Kang, J. R. Raney, P. Wang, L. Fang, F. Candido, J. A. Lewis, and K. Bertoldi. Multistable architected materials for trapping elastic strain energy. *Advanced Materials*, 27(29):4296–4301, 2015.

[7] H. Xiu, R. Fancher, I. Frankel, P. Ziemke, M. Fermen-Coker, M. Begley, and N. Boechler. Minimizing finite viscosity enhances relative kinetic energy absorption in bistable mechanical metamaterials but only with sufficiently fine discretization: a nonlinear dynamical size effect. *Journal of the Mechanics and Physics of Solids*, 200:106105, 2025.

[8] S. H. Strogatz. *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*. CRC Press, 2024.

[9] R. Fancher, I. Frankel, K. C. H. Chin, M. Abi Ghanem, B. MacNider, L. S. Shannahan, J. F. Berry, M. Fermen-Coker, A. J. Boydston, and N. Boechler. Dependence of the kinetic energy absorption capacity of bistable mechanical metamaterials on impactor mass and velocity. *Extreme Mechanics Letters*, 63:102044, 2023.

[10] P. W. Anderson. Absence of diffusion in certain random lattices. *Physical Review*, 109(5):1492–1505, 1958.

[11] M. I. Hussein, M. J. Leamy, and M. Ruzzene. Dynamics of phononic materials and structures: historical origins, recent progress, and future outlook. *Applied Mechanics Reviews*, 66(4):040802, 2014.

[12] T. V. Laptyeva, M. V. Ivanchenko, and S. Flach. Nonlinear lattice waves in heterogeneous media. *Journal of Physics A: Mathematical and Theoretical*, 47(49):493001, 2014.

[13] B. MacNider, H. Xiu, C. Tamur, K. Qian, I. Frankel, M. Brandy, H. A. Kim, and N. Boechler. Customizable wave tailoring nonlinear materials enabled by bilevel inverse design. *Nature Communications*, 16(1):3425, 2025.

[14] J. R. Dormand and P. J. Prince. A family of embedded Runge–Kutta formulae. *Journal of Computational and Applied Mathematics*, 6(1):19–26, 1980.

[15] L. F. Shampine and M. W. Reichelt. The MATLAB ODE suite. *SIAM Journal on Scientific Computing*, 18(1):1–22, 1997.

[16] C. E. Rasmussen and C. K. I. Williams. *Gaussian Processes for Machine Learning*. MIT Press, 2006.

[17] D. R. Jones, M. Schonlau, and W. J. Welch. Efficient global optimization of expensive black-box functions. *Journal of Global Optimization*, 13(4):455–492, 1998.

[18] P. Virtanen *et al.* SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods*, 17:261–272, 2020.

[19] S. K. Lam, A. Pitrou, and S. Seibert. Numba: a LLVM-based Python JIT compiler. In *Proc. Second Workshop on the LLVM Compiler Infrastructure in HPC*, 2015.